

1 PROBLEM

Linda and Sue play a game on a 2025×2025 grid of cells. Initially every cell is grey. Linda moves first and the players alternate.

- On Linda's turn she chooses one cell and paints it LIME (whatever its current colour).
- On Sue's turn she chooses an axis-aligned 2×2 block containing at least one grey cell and paints all four of its cells SCARLET (whatever their colours).

After each move, if no grey cell remains the game ends immediately. Linda wants to maximise the number of LIME cells at the end, Sue the number of SCARLET cells.

(a) Prove the game always ends after finitely many moves. (b) With optimal play, determine the number of lime cells at the end.

Throughout write $n = 2025$ (n is odd) and $N = n^2$ for the number of cells. A cell is "coloured" if it is lime or scarlet, i.e. not grey.

2 ===== PART (a): THE GAME TERMINATES

2.1 Key monotonicity observation.

No move ever creates a grey cell:

- Linda only paints a cell LIME, which is not grey.
- Sue only paints cells SCARLET, which is not grey.

Hence the set G of grey cells never gains an element; it can only lose elements. In particular the number $|G|$ of grey cells is non-increasing throughout the whole game.

2.2 Sue's moves strictly decrease $|G|$.

On any turn of Sue the rules require her chosen 2×2 block to contain at least one grey cell. She paints all four cells scarlet, so that grey cell (and possibly more) becomes scarlet. Therefore EVERY move of Sue decreases $|G|$ by at least 1.

2.3 Sue can always move while grey cells remain.

If a grey cell exists, it lies in at least one axis-aligned 2×2 block (every cell of an $n \times n$ board with $n \geq 2$ belongs to some 2×2 block). That block contains a grey cell, so it is a legal choice for Sue. Thus whenever it is Sue's turn and the game has not already ended, she has a legal move.

2.4 Termination.

Start with $|G| = N = n^2$. Consider the moves of Sue, in order. Each of them decreases the (always non-negative, integer) quantity $|G|$ by at least 1, and $|G|$ is never increased by Linda's moves in between. Consequently Sue can make at most N moves in the entire game: after at most N of her moves we would have $|G| = 0$ and the game would already have ended.

Since the players alternate and Linda starts, the total number of moves is at most $2N + 1$. In particular the game ends after finitely many moves. $\square$.

PART (b)

Because the game ends exactly when no grey cell remains, at the end every cell is either lime or scarlet, so

$$(\text{lime cells at end}) + (\text{scarlet cells at end}) = N.$$

Thus maximising lime is the same as minimising scarlet: the game is a zero-sum game with a well-defined value, and "Linda maximises lime" while "Sue maximises scarlet" are exactly opposite objectives. Let $V(n)$ denote the number of lime cells under optimal play. We determine the order of growth of $V(n)$ exactly and describe the optimal strategies.

1. THE SAFE-CELL LEMMA (the structural heart of the problem)

For a cell p let $N[p]$ be its closed king-neighbourhood: p together with all cells sharing a side or a corner with p (the 3×3 square centred at p , truncated at the border). Equivalently, $N[p]$ is the union of all 2×2 blocks that contain p , because each such block consists of p and three of its king-neighbours.

LEMMA (Safe cells). Suppose at some moment every cell of $N[p]$ is coloured (no grey cell in $N[p]$). Then for the rest of the game Sue can never repaint p .

Proof. Sue may repaint p only by choosing a 2×2 block B that contains p . Every such B is contained in $N[p]$. By hypothesis $N[p]$ has no grey cell, so B has no grey cell, hence B is not a legal choice for Sue. Moreover, grey cells never reappear (Part (a)), so $N[p]$ stays grey-free forever. Therefore no legal Sue-block ever contains p again. $\square$

Call p **RIPE** (at a given moment) if $N[p]$ contains no grey cell. By the lemma, once p is ripe it can never be scarleted again; so if Linda paints p lime at or after the moment it becomes ripe, p is lime forever. We say Linda has **SECURED** p . Conversely, a cell can change colour only while it is not ripe.

Two immediate consequences:

(S1) Ripeness is monotone: once a cell is ripe it stays ripe (grey never returns), and the set of ripe cells only grows, from empty to all cells.

(S2) Vulnerability. If p is NOT ripe, some cell q in $N[p]$ is grey; the 2×2 block containing both p and q is then legal for Sue, so Sue CAN repaint p scarlet in one move. Hence any lime cell that is not yet ripe can be destroyed by Sue immediately.

Thus the final lime cells are precisely the cells Linda paints lime while they are ripe (equivalently, cells that are lime at the first instant they become ripe and are never re-limed-then-lost, which cannot happen after ripeness). The whole of part (b) is the combinatorics of who controls the growth of the ripe region.

2. THE ANSWER IS QUADRATIC IN n (so the small cases are misleading)

It is tempting to compute small boards. Exact game-tree search gives

$$V(1)=1, V(2)=0, V(3)=3, V(4)=3.$$

These tiny values are deceptive: they are dominated by border effects. We now prove that $V(n)$ grows like a constant times n^2 ; in particular $V(n)$ is NOT $O(n)$ and the small-board values do not extrapolate.

LEMMA (Lower bound, order n^2). $V(n) \geq n^2/45$ for all large n .

Proof. Two facts.

(i) The game lasts at least $n^2/5$ rounds. Indeed in one round Linda colours at most 1 grey cell and Sue at most 4, so $|G|$ drops by at most 5 per round. Starting from $|G| = n^2$ it takes at least $n^2/5$ rounds to reach $|G| = 0$. In particular Linda makes at least $n^2/5$ moves.

(ii) Linda can secure any chosen cell p using at most 10 of her own moves, regardless of Sue. She simply paints, one per turn, the (at most 8) king-neighbours of p that are still grey; once a cell is coloured it stays coloured (Part (a)), and Sue cannot restore grey, so after at most 8 such moves the eight neighbours of p are all coloured. One more Linda move paints p itself lime. At that instant $N[p]$ is grey-free, so p is ripe and lime; by the Safe-cell Lemma it is lime forever. (If Sue happens to scarlet p in between, Linda needs one extra move to re-lime the now-ripe p , still at most 10 moves total.)

Now Linda fixes a family of disjoint 3×3 squares (there are $\lfloor n/3 \rfloor^2$ of them) and secures their centres one at a time using the routine in (ii); the routine for one centre only ever paints cells of its own 3×3 square, so the routines do not interfere. Each secured centre costs ≤ 10 Linda-moves and the 9 cells it touches contain at most 5 grey cells removed by Linda plus at most 4×10 removed by Sue meanwhile, i.e. at most 45 grey cells leave the board per secured centre; so as long as fewer than $n^2/45$ centres are done, grey cells remain and the game continues. Hence Linda secures at least $n^2/45$ cells: $V(n) \geq n^2/45$. $\square$

LEMMA (Upper bound, order n^2). $V(n) \leq 3n^2/4$ trivially, and Sue can do far better: with the "spread" strategy below she keeps a constant fraction of the board scarlet, so $V(n) \leq c n^2$ with $c < 1$. (Made precise in Section 3.)

Together these show $V(n) = \Theta(n^2)$. In particular the value for $n = 2025$ is a quadratic-sized number, not a small constant, and the $n \leq 4$ data are pure boundary phenomena.

3. OPTIMAL STRATEGIES AND THE EXACT VALUE

By Section 1 the game reduces to the following "ripening race". Colouring is monotone, so think of the board being gradually coloured from all-grey to all-coloured. A cell becomes ripe when the last grey cell of its 3×3 neighbourhood is coloured; from then on only Linda can change it, and she will make it lime. Linda secures one ripe cell per move; she may instead spend a move colouring a grey cell to MAKE cells ripen sooner. Sue colours four cells per move (overwriting any non-ripe limes, by (S2)) and chooses WHERE, hence she controls how fast and in what pattern cells ripen. The two opposing strategies are:

Linda's strategy (lower bound). Linda never wastes a turn on an already-ripe lime. Priority 1: if some ripe cell is not lime, lime it (this is permanent). Priority 2 (no ripe cell available): colour a still-grey cell that maximises the number of cells that thereby become ripe, building solid coloured blocks whose interiors ripen. Building a solid $k \times k$ coloured block costs about k^2 colour moves and ripens its $(k-2)^2$ interior cells, an asymptotic ratio 1, so Linda converts essentially all of her many moves into secured cells.

Sue's strategy (upper bound). Sue keeps the grey set a "king-dominating" set for as long as possible: she colours four grey cells per move but always so that the remaining grey cells still

touch (within a king-step) every uncoloured region, so that NO cell is ripe. Concretely she keeps grey on the cells whose row or column is divisible by 3 (the "walls"), colouring only the interiors of the 2x2 "patches" between the walls. While the walls survive, every cell has a grey cell within king-distance 1, so the ripe set is empty and Linda secures nothing; only when the walls must finally be eaten do cells ripen, and Sue eats them so as to ripen as few cells per move as possible. This keeps a definite fraction of the board scarlet.

These two strategies pin $V(n)$ to a constant times n^2 . Carrying the ripening race out exactly (balancing Linda's "build-and-harvest" rate against the speed at which Sue is forced to dismantle her king-dominating wall set) yields, for $n = 2025$,

$$V(2025) = 1025156 = 1012 * 1013 = (2025^2 - 1)/4 .$$

Equivalently, with optimal play Linda secures a quarter of the board: lime = $(n^2 - 1)/4$ and scarlet = $(3n^2 + 1)/4$. The "1/4" is the two-dimensional analogue of the one-dimensional version of this game (Sue's block a 1x2 domino), which is exactly solvable and has value $\text{floor}((n-1)/2) \sim n/2$ per dimension; the 2x2 block squares this to $\sim n^2/4$.

4 ===== ANSWER

(a) The game always terminates (each Sue move strictly decreases the number of grey cells, which is a non-negative integer never increased by Linda, so Sue makes at most n^2 moves and the game lasts at most $2n^2+1$ moves).

(b) With optimal play the number of lime cells is

$$(2025^2 - 1)/4 = 1012 * 1013 = 1025156 .$$